

Dhanush Kumar S V

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SUMMARY

Process Engineer skilled in process simulation, optimization, and manufacturing improvement. Experienced in using Aspen Plus, Aspen HYSYS, and MATLAB for modeling chemical and semiconductor processes. Strong understanding of yield improvement, defect reduction, and process integration. Dedicated to enhancing production efficiency and product quality through data-driven analysis and continuous improvement.

EDUCATION

National Chung Hsing University, Taichung, Taiwan

M.S. in Chemical Engineering (2024–Present) GPA: 3.95/4.3

Anna University, Chennai, India

B.Tech in Chemical Engineering (2019–2023) GPA: 7.84/10

TECHNICAL EXPERTISE

Process Simulation: Aspen Plus, Aspen HYSYS, MATLAB, GAMS, Ansys (Fluent)

Process Integration: Electrochemical coating, surface process modeling, yield optimization

Data Analysis: Origin, Excel, ImageJ, Python

Documentation: Process flow reports, design of experiments, technical writing

RESEARCH EXPERIENCE

Hybrid VMD–MED / MSF–MED Process Modeling for Phosphogypsum Wastewater Treatment

- Modeled hybrid VMD–MED and MSF–MED systems for phosphogypsum wastewater purification using process simulation.
- Integrated microbial electrochemical cell (MEC) for enhanced phosphorus recovery and energy-efficient treatment.
- Performed process optimization and energy analysis to improve water recovery, phosphorus yield, and overall system efficiency.

MILP-Based Optimization of Cooperative Dairy Milk Supply Chain

- Formulated a GAMS-based MILP model for integrated village–BMC–plant milk procurement under capacity, flow balance, and service constraints.
- Conducted sensitivity and scenario analyses on BMC capacity and penalty parameters to evaluate cost–service trade-offs.

- Achieved ~17% total cost reduction, up to ~30% decrease in unserved milk, and identified key capacity thresholds.

Hydrogen Production using the Photocatalytic Method

- Designed and built a 4 L trapezoidal photocatalytic reactor for solar-driven hydrogen production from sulphuric wastewater.
 - Achieved $\sim 300 \text{ mL h}^{-1} \text{ L}^{-1}$ hydrogen production using TiO_2 photocatalyst under direct sunlight.
 - Optimized catalyst loading and sulphide concentration to maximize hydrogen yield and process efficiency.
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INTERNSHIPS

University Teaching Assistant - National Chung Hsing University, Taiwan (Sep 2025 – Dec 2025)

- Guided students in process simulation using Aspen Plus, troubleshooting modeling and convergence issues.
- Supported design and optimization of chemical processes through hands-on simulation exercises.
- Assisted in teaching optimization concepts using GAMS, including constraints and objective functions.
- Developed structured problem-solving approaches for process design and analysis.

Quality Assurance Trainee – Steel Authority of India Limited (SAIL), Salem (Oct 2022 – Nov 2022)

- Analyzed production data from hot and cold rolling lines to identify factors affecting steel yield and quality.
- Investigated manufacturing defects and contributed to root cause analysis for process improvement.

CFD Research Intern – Indian Institute of Technology, Indore (Nov 2021 – Jan 2022)

- Performed computational fluid dynamics analysis for chemical processes, learning parameter optimization and reporting techniques.

Surface Coating Intern – RK Metals (Aug 2021)

- Studied electrochemical coating and metallization methods used in industrial surface treatment.
 - Evaluated coating performance factors including thickness, adhesion, and corrosion resistance.
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PUBLICATIONS & CONFERENCES

- “Location Selection and Purification Process Simulation for a Glycerol Plant.” *ICATES 2024*
 - “Production of Hydrogen Gas Using Photo-catalytic Method.” *ICATS 2023*
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LANGUAGES

English (Fluent) | Chinese (Basic) | Tamil (Native)